

Cloudflare D1 Write Audit Report

Database ailatest-journal (uuid 861dd053-c23a-4363-b62b-2559c06ec3fd) · Account e7dbf2ec3ab60aa5557cc2364a7a2f26 · Source: Cloudflare GraphQL API (d1AnalyticsAdaptiveGroups)

Defect introduced
2026-06-13

Steady writes began
2026-06-21

Upgraded to Paid
2026-08-10

Fixed
2026-08-20

Peak daily writes
22.6M rows
(08-10)

Daily D1 rows written (2026-06-13 → 08-20)



Root cause (three stacked code defects)

- ① Over-frequent scheduled job: cron set to `*/15 * * * *` (every 15 minutes).
- ② Over-wide backfill window: each run re-scanned **the last 90 days** of event data for traffic classification.
- ③ UPDATE with no dedup filter: the UPDATE only filtered by time range, not by "already classified", so already-classified rows were rewritten every run. D1 bills UPDATES per row even when the value is unchanged.

Fixes deployed (2026-08-20, same day)

- ✓ Disabled the high-frequency cron, keeping only a once-daily `12 16 * * *`
- ✓ Narrowed the backfill window from 90 days to 1 day

- ✓ Added AND (traffic_type='' OR visitor_hash='') so the UPDATE only touches unclassified rows

Daily write details

Date	Rows written	Est. cost (\$)
06-13	2,454,306	2.45
06-14	1,566,333	1.57
06-15	1,668,490	1.67
06-16	8,430	0.01
06-17	15,627	0.02
06-18	13,949	0.01
06-19	9,322	0.01
06-20	5,126	0.01
06-21	4,858,868	4.86
06-22	4,957,126	4.96
06-23	5,741,505	5.74
06-24	5,154,416	5.15
06-25	6,064,872	6.06
06-26	7,548,369	7.55
06-27	7,960,850	7.96
06-28	7,313,129	7.31
06-29	8,048,023	8.05
06-30	8,706,314	8.71
07-01	8,048,299	8.05
07-02	8,713,388	8.71
07-03	9,644,849	9.64
07-04	9,825,224	9.83
07-05	8,743,312	8.74
07-06	8,503,203	8.50
07-07	8,610,987	8.61
07-08	8,592,763	8.59

Date	Rows written	Est. cost (\$)
07-09	9,902,167	9.90
07-10	8,723,698	8.72
07-11	8,284,306	8.28
07-12	8,570,165	8.57
07-13	8,757,149	8.76
07-14	9,914,936	9.91
07-15	10,490,728	10.49
07-16	10,690,356	10.69
07-17	10,542,775	10.54
07-18	9,886,293	9.89
07-19	12,524,074	12.52
07-20	11,258,584	11.26
07-21	11,911,240	11.91
07-22	11,573,240	11.57
07-23	11,488,821	11.49
07-24	11,337,094	11.34
07-25	10,863,056	10.86
07-26	11,812,922	11.81
07-27	10,971,809	10.97
07-28	11,413,747	11.41
07-29	10,234,403	10.23
07-30	12,883,401	12.88
07-31	12,657,611	12.66
08-01	12,911,323	12.91
08-02	12,520,958	12.52
08-03	13,132,377	13.13
08-04	11,318,386	11.32
08-05	10,389,383	10.39
08-06	19,924,494	19.92
08-07	14,688,686	14.69

Date	Rows written	Est. cost (\$)
08-08	17,730,361	17.73
08-09	11,797,588	11.80
08-10	22,574,597	22.57
08-11	14,520,150	14.52
08-12	13,613,532	13.61
08-13	13,667,403	13.67
08-14	13,710,246	13.71
08-15	14,241,982	14.24
08-16	14,084,543	14.08
08-17	14,103,960	14.10
08-18	14,125,453	14.13
08-19	13,953,391	13.95
08-20	4,745,444	4.75

Note: cost is estimated per D1 pricing (\$1 / million rows beyond the 50M monthly free allowance). 08-20 is the fix day, data through 16:32 UTC.